

# Storage Devices

Storage devices are the computer hardware used to remember/store/back up data. There are many types of storage devices, each with their own benefits and drawbacks. Below are explanations about three different storage devices.

## 1. Magnetic Storage Devices.

Hard Disk Drive (HDD)



Internal Hard disk



External Hard



### a) What is a hard disk drive?

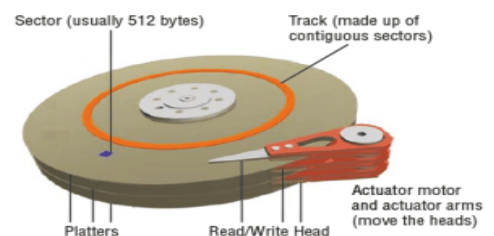
Hard disk drives are non-volatile magnetic storage devices capable of remembering vast amounts of data. An electromagnet in the read/write head charges the disk's surface with either a positive or negative charge, this is how binary 1 or 0 is represented. The read/write head is then capable of detecting the magnetic charges left on the disk's surface, this is how data is read. The disk surface is divided into concentric circles (tracks) and sectors (wedges). Dividing the surface in this way provides physical addresses to remember where data is saved. A circuit board carefully co-ordinates the rotating disk and swinging actuator arm to allow the read/write head to access any location very quickly. Typical HDD capacities are measured in Terabytes (TB). They can be installed inside a computer or purchased in a portable (external) format.

### Typical applications for hard disk drives

- Desktop computers
- Laptop computers
- TV and satellite recorders
- Servers and mainframes
- Portable (external) drives are sometimes used to back up home computers or transfer large files

### Benefits of hard disk drives

- Capable of holding vast amounts of data at affordable prices
- Fast read and write speeds

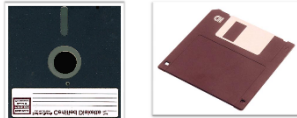


- Reliable technology
- Relatively small in size

### **Drawbacks of hard disk drives**

- Due to the nature of its moving parts, they will eventually wear and break.
- Very fragile and sensitive to shocks.
- High power consumption.
- Some noise is created by the moving parts

### **b) Floppy Disk**



Also known as a diskette, floppy, or FD, the floppy disk is another type of storage medium that uses magnetic storage technology to store information. Floppy disks were once a common storage device for computers and lasted from the mid-1970's through to the start of the 21st century. The earliest floppies were 8-inch (203 mm) in size, but these were replaced by 5 1/4-inch (133 mm) disk drives, and finally a 3 1/2 inch (90 mm) version.

### **c) Tape**

In the past, magnetic tape was often used for digital data storage, because of its low cost and ability to store large amounts of data. The technology essentially consisted of a magnetically thin coated piece of plastic wrapped around wheels. Its relative slowness and unreliability compared to other data storage solutions has resulted in it now being largely abandoned as a media

## **2. Optical Storage**

### **CD, DVD and Blu-Ray Discs**



#### **What are optical storage discs?**

CD, DVD and Blu-Ray drives are optical storage devices. Binary data is stored as changes to the texture of the disc's surface, sometimes thought of as microscopic pits and bumps. These 'bumps' are located on a continuous spiral track, starting at the centre of the disc. Whilst the disc is rotating at a constant speed, a laser is pointed at the spiral track of 'bumps'. The laser will reflect/bounce off the disc surface in different directions depending upon whether a 1 or 0 has been read.

### **Disc capacities**

In the pursuit of larger optical storage capacities, DVDs were created, followed by Blu-Ray.

| CD     | DVD    | Blu-Ray        |
|--------|--------|----------------|
| 700 MB | 4.7 GB | 25 GB – 128 GB |

### Typical applications for optical media

- CD – Audio and small amounts of data
- DVD – Standard definition movies and data
- Blu-Ray – HD video and large amounts of data

#### DVD

Despite being the same physical size, a DVD can hold more data than a CD. To achieve this, a more tightly packed spiral track is used to store the data on the disc. To accurately access the smaller 'bumps', a finer red laser is used in a DVD drive than that found in a standard CD drive. To increase capacity further, DVDs are also capable of dual layering.

#### Blu-Ray

Blu-Ray technology squashes even more data into the same size disc as a CD or DVD. The spiral data tracks on a Blu-Ray disc are so small that a special blue-violet laser, hence its name, has to be used to read the 'bumps'. Like a DVD, Blu-Ray discs are capable of storing data on multiple layers.

### Recordable Optical Media:

- **CD-ROM, DVD-ROM, Blu-Ray-ROM**

*Read only* – the data is permanently written to the disc at the point of manufacture.

- **CD-R, DVD-R, BD-R**

*Recordable* – blank discs that can be burnt (written to) once.

- **CD-RW, DVD-RW, BD-RE**

*Re-writable* – blank discs that can be burnt (written to) over and over again (can be erased and reused many times).

### 3. Flash Memory

#### USB Flash Memory



#### What is USB Flash Memory?

Flash are non-volatile solid state storage devices which use NAND flash memories to store data (millions of transistors). USB refers to the USB connection that allows users to plug the device into the USB port of a computer. Other types of flash storage include the memory cards used in digital cameras. Flash memory comes in a variety of capacities to suit most budgets and requirements.

#### Typical applications for flash memory

- USB memory sticks – saving and transferring documents etc
- Memory cards in digital cameras

#### Benefits of flash memory

- Portable, lightweight and small enough to fit a key ring.
- Durability, flash has no moving parts to damage
- Range of capacities available.
- Fast speeds, with no moving parts of boot up time

#### Drawbacks of flash memory

- Limited (but huge) number of write cycles possible
- Really high capacities are uncommon
- In relative terms, an expensive storage option compared to a HDD

### Workshop

#### 1) Insert the suitable words from the box

|                                                       |
|-------------------------------------------------------|
| Magnetic, disc, disk, sized, DVD, formats, CD, output |
|-------------------------------------------------------|

In later optical disc systems, a laser read the dots and the data was converted to an electrical signal. Then the data was converted to audio or visual ..... However, the technology didn't appear in the marketplace until Philips and Sony came out with the compact disc , ....., in 1982. Since then, there has been a constant

succession of optical disc ....., first in CD formats and followed by a number of digital versatile disc , ....., formats. Initially, an optical disc could hold much more data than similarly- ..... magnetic storage media. Increases in hard ..... drive (HDD) technology, and particularly the introduction of solid-state memory, helped magnetic media capacity outgrow optical discs. However, optical ..... storage is less likely to degrade over time than ..... storage, such as HDD or magnetic tape, so it is often used for archival or cold storage.

## 2) Choose the appropriate option .

Flash memory is a type of non-volatile memory that can be electronically erased and ( re.../ inter.../multi...) ..... programmed. Its name was invented by Toshiba to express how ( many, much, less) ..... faster it could be erased ( ? / ! / -) ..... 'in a flash' which means very quickly . Unlike RAM which is (non-volatile, versatile, volatile ) ....., flash memory retains the information stored in the chip when the power is turned off. This makes it ideal for the use in ( digit, digital, digitally) ..... cameras, laptops, video game cards, portable media and mobile phones. In addition, it offers ( fastly, fast, faster) ..... read access with transfer rates of 12MB per second. Unlike ROM chips , flash memory chips are (rewritable, read only, recordable) ..... , so you can update data via software .

## 3) Insert the suitable verbs or modals to complete the tips about how to protect your hard drive .

*Stored, format, repair, wait, shouldn't, known, can, don't, erases.*

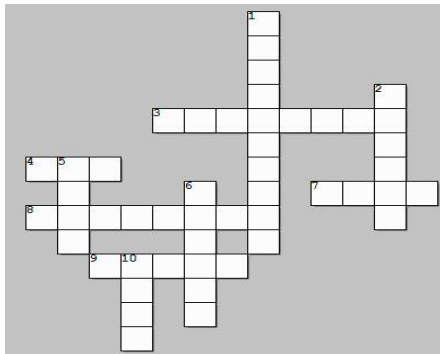
\*..... hit or move the computer while the hard drive is spinning. Hard drives are very sensitive to vibration and shocks, especially when they are operating ; when the read and write head touches the rotating disk , it ..... scratch and damage the disk surface. This is .....as **head crash** .

\*You..... turn your computer off and on quickly. .... at least ten seconds to ensure that the drive has stopped spinning

\*to check and ..... a drive, you can use a disk diagnosis utility like Windows ScanDisk.

\*Formatting ..... any existing files on a disk, so do not ..... disks on which data that you don't want to lose is..... . You should also back up your hard drive regularly.

4) This crossword puzzle is based on the definition and examples of magnetic storage. For this activity, complete the crossword by filling in a word that fits each of the given clues.



## Across

3. A \_\_\_\_\_ hard drive is a portable storage device that is usually in a metal or plastic case and can be connected to a computer via USB or wirelessly.

4. The equivalent of a floppy disk in this modern time.

7. The trends in storing data of a computer system, such as the invention of the harddisk drive, has caused the large decline in the use of \_\_\_\_\_ backup systems.

8. RAM is classified as a \_\_\_\_\_ storage because it requires constant power supply to preserve stored data.

9. Magnetic Storage is highly preferred because it is \_\_\_\_\_ compared to other storage systems.

## Down

1. A magnetic storage refers to data storage using a \_\_\_\_\_ medium to store digital data.

2. This made use of a thin disk and flexible magnetic storage media, which provided efficient ways to move small amounts of data between computers.

5. A limitation of a hard disk drive is in accessing the stored data, which is rather \_\_\_\_\_ when compared to a solid state drive.

6. This is the type of information that makes up the digital data whose unit can take on only two possible states: either zero or one.

10. The \_\_\_\_\_ is a component of a magnetic storage device that detects and modifies the magnetization of the material.